Can an LLM extract meaningful trading signals from SEC filings?

FilingSignal, spec & summary · Final project: AI & Innovation in Capital Markets · Track 3: AI application for investors

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Code: github.com/aviad59/algo-trade

1. The question

Every quarter, thousands of companies file hundreds of pages each with the SEC. The *historical* content of those filings, revenue and balance sheet, is priced within minutes. What is not priced instantly is the **forward-looking layer**: capex plans, offtake agreements, mine expansions and restarts, guidance on volumes. That layer is unstructured prose, scattered across dozens of companies, and reading all of it is exactly the kind of work nobody does at scale.

So the question this project puts to a real test is narrow and falsifiable: **can a large language model read EDGAR filings, extract genuine forward-looking signals, and turn them into a market forecast that survives an honest backtest?** Not "can an LLM talk about stocks", but can it produce something measurable that beats the alternatives you could actually trade.

2. What FilingSignal is

FilingSignal is two things in one codebase, and the pairing is the point:

- an **interactive web application**: a four-page React dashboard where an investor sees the current quarter's ranking, the filings and quotes behind it, and how every past prediction actually performed;
- a **research platform**: a point-in-time scoring engine, an evaluation suite, and a backtest harness deliberately built to be able to *reject* the hypothesis.

The governing design rule is **"math rates, the LLM explains."** The language model never ranks anything and never sees a price. It only converts filing prose into structured, quoted records. A deterministic scorer, pure arithmetic with no LLM in it, turns those records into the ranking. That is what separates the project from a chatbot that opines about stocks: every ranking is reproducible from the database, and every input traces back to a verbatim sentence in a real filing.

3. The pipeline, five stages

A filing enters as prose and leaves as a rank. Only stages 3 and 4 touch a language model, and only stage 3 reads the filing itself. Each stage writes to the next through the SQLite buffer, so the expensive step runs once and the cheap steps can be re-run forever.

Stage	What happens	Why it matters
1 · Ingest	Fetch filings from EDGAR, isolate the relevant sections (MD&A, risk factors, selected 8-K items), condense to forward-looking text.	A full 10-K is far too large and mostly boilerplate. Condensing cuts tokens by about 5.7x.
2 · Filter	A two-part gate: allowed forms and items, then a material keyword test.	Routine filings are rejected *before* any spend, and already-digested filings are skipped.
3 · Extract	**Agent #1** returns structured records only. Each effect carries a date window, a confidence, and a verbatim quote.	The quote is the anti-hallucination anchor: no textual anchor, no effect.
4 · Score	Deterministic scoring and ranking. **Agent #2** writes the human-readable rationale afterwards.	The decision is reproducible and auditable; the prose sits downstream of it, never upstream.
5 · Evaluate	Rank-IC with a permutation test, plus a four-baseline backtest.	Return alone can be luck. This stage is what makes the answer trustworthy, or negative.

4. The universe: producers, consumers, and the forms we read

The experiment is scoped to **raw materials**, one of the few corners of the market where the supply chain is *explicitly written down*: a miner states added tonnes and a schedule, a manufacturer states demand. Each material is frozen to a liquid miner-equity ETF, so a forecast converts directly into a tradeable position.

Every company carries a **perspective**. **Producers** are the supply side (what is being dug up and when), **consumers** the demand side (who is buying it and for what). Each material therefore gets evidence from two independent directions, which lets the scorer ask whether unrelated companies agree instead of trusting one voice.

Material	ETF	Producers (supply side)	Consumers (demand side)
Copper	`COPX`	FCX, GOLD, HBM, RIO, SCCO, TECK	EMR, ETN, TSLA
Steel	`SLX`	CLF, NUE, RIO, STLD, TECK	CAT, DE, F, GM
Rare Earths	`REMX`	MP, USAR, UUUU	BA, LMT, NOC
Gold	`GDX`	AEM, CDE, FCX, GOLD, KGC, NEM, WPM	none: producer-only, demand is macro and not filed
Uranium	`URNM`	CCJ, LEU, NXE, UEC, UUUU	none: utilities file too little to be usable
Silver	`SIL`	CDE, HL, PAAS, SCCO, WPM	none: producer-only

33 companies, several spanning two materials (Freeport files on copper and gold). The map is frozen in `universe/materials.yaml`, fixed *before* the backtest ran.

Which filings are read

Companies split into two tiers, each with its own form set and its own extraction prompt. A 10-K, a 10-Q and an 8-K carry different time horizons and different meaning for the same sentence.

Tier	Companies	Forms read	Character of the filing
US	24	`10-K` annual, `10-Q` quarterly, `8-K` event	Strategic outlook, quarterly update, discrete material events
Foreign	9	`40-F`, `20-F`, `6-K`	Canadian and other foreign issuers, the equivalents of the above

Corpus as submitted: **577 filings** digested, yielding **1,102 dated effects**, 590 extraction records and 740 recorded warnings.

5. What the system predicts

The output is deliberately shaped like a decision, not a commentary:

> **Given every filing that was public before this quarter began, rank the six materials, and if I had to invest in exactly one of them for the coming quarter, which one is it?**

That question has a single answer and is measurable in hindsight. The ranking is produced as follows, for a target quarter **Q** whose **decision date is its first day**:

```
contribution(e) = sign * magnitude * confidence * recency * overlap
  sign       = +1 (increase) | -1 (decrease)           magnitude  = {small:1, moderate:2, large:3}
  confidence = extractor confidence in [0,1]           recency    = exp(-ln2/12 * age_months)  # 12-month half-life
  overlap    = |effect window ∩ Q| / |Q|                                                    # share of Q covered

producer_score = producer_raw * (1 - exp(-n_p / k))     # breadth gate, k = 2 ; n = distinct tickers contributing
consumer_score = consumer_raw * (1 - exp(-n_c / k))
combined(m,Q)  = producer_score + consumer_score        z(m,Q) = (combined - mean) / std  # cross-sectional
pick(Q)        = argmax_m z(m,Q)
```

Two pieces carry the intuition. **Recency decay** means a year-old statement counts half as much as a fresh one. The **breadth gate** encodes "one company is noise": the score stays suppressed while a single source is talking and opens up as more independent companies say the same thing. Standardizing across the six converts the question from "is copper positive" into "is copper positive *relative to the other five*", which is the actual choice being made.

The hyperparameters are **modeling choices, not truths**: sensible defaults set up front and explicitly **never tuned on the test period**, since tuning them there would empty the backtest of meaning.

6. The application: four pages

The dashboard is a React 19 and TypeScript SPA with no third-party routing, charting or state-management libraries. It reads a read-only API, and the whole stack ships as a single Docker image serving both the SPA and `/api/v1` on one port.

Home: what the machine actually does

An interactive walkthrough of the five-stage pipeline. Each stage can be opened to show what enters it and what leaves it, so a reader understands where the LLM is used and, just as importantly, where it is *not*, before looking at a single number.

Forecast: the current quarter's call

The top pick for the live quarter, with its ETF, z-score, margin over the runner-up, and a rank-confidence label. Below it sit the full six-material ranking table with producer and consumer sub-scores and filing counts, a diverging bar chart of this quarter's z-scores, a six-quarter score-trend chart, Agent #2's written rationale for the pick, and the verbatim filing quotes it cited. A live tracker card shows how the current call is doing quarter-to-date: the pick's return, its standing against the baselines, and a provisional rank-IC.

Filings: the evidence layer, and a live extraction

Every digested filing with its summary and its extracted effects: material, perspective, direction, magnitude, date window, confidence, and the quote that justifies each one, plus counters for filings, extractions, effects and average confidence. Any effect can be traced back to the sentence that produced it.

This page also hosts the **live extraction demo**. Choose a ticker, a form and a date, and the app fetches that single filing from EDGAR and runs **Agent #1 on it in real time**, returning the summary and the extracted effects on screen. It is the one place a reviewer can watch the AI feature work end to end on a filing of their own choosing. The endpoint is gated by a shared access key and returns `503` when no key is configured, so a public deployment cannot be made to burn the provider key.

Backtest: the verdict

A quarter picker walks through history. For each quarter it shows the **predicted ranking as it stood on day one of that quarter**, what actually happened, the pick's return against the baselines, prediction-vs-realized slopes, and the part that makes it auditable: **the filings that drove that specific pick**. Across the full window it shows the $100 equity curve against all four baselines, the metric tiles, a per-quarter rank-IC chart, a hit grid, and the producer-vs-consumer decomposition.

7. The forecast is live, not a snapshot

The Forecast page is not a frozen demo of a past quarter. It tracks the **real, in-progress quarter**: the ranking was computed from filings public before the quarter began, and the tracker updates against actual ETF prices as the quarter unfolds. The number on screen is a standing prediction that has not yet been resolved.

The rollover mechanism

The whole system derives its state from one clock seam, `quarter_of(today())`, so the calendar rather than a human decides what "the current quarter" means:

- **End-of-quarter sweep.** `filingsignal refresh` walks the universe incrementally: it skips filings already digested, drops routine ones at the pre-LLM filter, and spends a model call only on genuinely new filings. It aborts immediately if the provider key is missing or exhausted, so there is no runaway spend.
- **Automatic re-anchoring.** The moment the calendar turns, say on Oct 1, the forecast quarter moves from Q3 to Q4 on its own. The quarter label, the published date, the point-in-time cutoff, the filing count and the ranking all recompute with zero code edits.
- **Hand-off between pages.** The Forecast page owns the unfinished quarter. The moment that quarter closes it becomes one more point-in-time row on the Backtest page, so today's live call is tomorrow's backtest evidence, under the same cutoff rule and with no retro-fitting.
- **Testable, not promised.** An environment override (`FILINGSIGNAL_TODAY`) moves the clock, so "what happens when the quarter turns" is covered by an automated test instead of a three-month wait.

8. The backtest: what exactly was tested

The backtest is the rollover mechanic replayed over history, so the question stays honest: *if I had run this every quarter using only what was public at the time, what would have happened?*

- **Point-in-time by construction.** The scorer takes the decision date as a parameter and structurally drops any effect whose filing became public after it. The 2025 Q4 ranking sees only filings through 2025-09-30. The live forecast and the backtest call the *same* function, so there is no look-ahead switch to forget to turn off.
- **The rule.** On the first day of each quarter, buy the top-ranked material's miner ETF, hold it for the quarter, rotate at the next quarter's open. 10 bps round-trip cost, dividend-adjusted closes, $100 starting capital.
- **Window.** 14 quarters, **2023 Q1 to 2026 Q2**, the span the filing corpus supports, chosen by data availability rather than by which quarters flattered the result.
- **Four baselines. SPY** (do nothing), **equal-weight** across all six materials (does picking add anything over naive diversification?), **random-pick Monte-Carlo** over 2,000 sequences (the luck floor, and the source of the return p-value), and **hindsight-best** (the ceiling that requires foresight).
- **A skill test, not just a return test.** Rank-IC is the Spearman correlation between the forecast ranking and the realized ranking, judged against a 2,000-shuffle permutation test. Return can be luck; rank-IC asks whether the *ordering* carried information.

9. Results: 14 quarters

Metric	FilingSignal	SPY	Equal-weight	Random (median)	Hindsight
Final value from $100	$281	$204	$201	$193	$1,332
CAGR	+34.4%	+22.7%	+22.0%	+20.6%	+109.5%
Sharpe	0.94	**1.85**	0.91	0.84	2.31
Max drawdown	-19.6%	**-4.5%**	-13.8%	-14.0%	-8.0%
Quarters beating equal-weight	7 / 14 (50%)	n/a	n/a	n/a	n/a

Significance: mean rank-IC **-0.21** over 12 scoreable quarters, and a 2,000-shuffle permutation gives **p = 0.95**. Return against 2,000 random-pick sequences gives **p = 0.15**. IC decomposition: producer-only -0.24, consumer-only -0.00, combined -0.21.

The decisions, quarter by quarter

Quarter	Pick	ETF	Realized	Equal-weight	Excess
2024 Q1	Uranium	URNM	+4.0%	+0.6%	+3.4
2024 Q2	Copper	COPX	+5.9%	-1.4%	+7.3
2024 Q3	Copper	COPX	+4.0%	+6.7%	-2.9
2024 Q4	Copper	COPX	-19.5%	-13.8%	-5.9
2025 Q1	Copper	COPX	+1.4%	+5.3%	-4.0
2025 Q2	Copper	COPX	+15.0%	+18.3%	-3.3
2025 Q3	**Rare Earths**	**REMX**	**+63.2%**	**+37.9%**	**+25.2**
2025 Q4	Rare Earths	REMX	+10.7%	+12.2%	-1.5
2026 Q1	Uranium	URNM	+4.5%	+7.6%	-3.2
2026 Q2	Copper	COPX	-1.2%	-8.6%	+7.3

This table shows the behaviour better than any summary statistic. The system stays on copper for five consecutive quarters, a consequence of slow recency decay and copper's high filing volume, and its win comes from catching the rotation into rare earths early in 2025 Q3, not from steady accuracy.

10. Conclusion

Over this 14-quarter window the strategy **beat every tradeable baseline on raw return**: $281 against SPY's $204, equal-weight's $201 and a random-pick median of $193. That is real, and it is not the whole answer:

- **Not statistically significant.** The return edge beats about 85% of random-pick sequences, so **p = 0.15**, above the conventional 0.05 threshold. A result like this arises by luck roughly one try in seven.
- **No cross-sectional skill.** The mean rank-IC is *negative* (**-0.21**, p = 0.95): the full ordering of the six materials carries no information, and whatever worked lived in the top pick alone.
- **Two quarters carry everything.** Remove 2025 Q3 (+63%) and 2023 Q3 (+41%) and $100 becomes **$122**, against $147 for equal-weight and $196 for SPY.
- **Risk-adjusted, it loses to doing nothing.** Sharpe 0.94 against SPY's 1.85, with a -19.6% drawdown against -4.5%. One high-beta miner ETF per quarter is far more volatile than the return line suggests.

The defensible claim: the top-1 rule beat its tradeable baselines over this window, driven by two large correct calls, with no evidence of ranking skill and no statistical significance.

So the answer to the opening question is a qualified *not yet*. Extracting structured, quoted, forward-looking claims from filings demonstrably works, across 577 filings and 1,102 effects. Turning them into a ranking that beats chance is what remains unproven. The contribution is the machinery that can tell the difference: structural point-in-time cutoffs, four baselines including a luck floor, and a permutation test allowed to return an unflattering number, which it did.

11. Limitations

- **Small sample:** 14 quarters, 6 assets, one sector, too few observations for significance except at extreme effect sizes.
- **Concentration:** two quarters produce the entire edge, and average metrics mislead on a fat-tailed profile.
- **No sector downturn tested:** the window overlaps a positive stretch for materials.
- **US-heavy:** foreign forms (40-F, 20-F, 6-K) extract thin, because section isolation there is not standardized.
- **The model is non-deterministic:** the quote constrains hallucination but not misinterpretation, and magnitude and date window are judgment calls.
- **Imperfect attribution chain:** a statement about a material need not move that material's ETF, since index composition, currency and macro intervene.
- **Costs are optimistic:** 10 bps excludes spread, slippage, tax and financing.
- **Full concentration:** one asset per quarter is not a retail-appropriate risk profile.

12. Future work

- **More sample, the top priority.** Extend the window backwards and widen the universe beyond one sector. Without it, no algorithmic improvement can be demonstrated.
- **Section isolation for foreign forms:** the largest coverage win available, and a direct fix for the US bias.
- **Hyperparameter sensitivity sweep:** a heat map over $\{\lambda, k, w_p, w_c\}$. Does the result survive a reasonable range, or depend on one point?
- **Portfolio instead of a bet:** a z-weighted or top-2 basket, or long/short of first against last. This uses the *whole* ranking rather than one cell, and cuts volatility.
- **Learn from the negative IC.** A persistently negative IC is itself information: a producer ramp may signal *oversupply* rather than demand, testable via a controlled sign flip out of sample.
- **Shorter horizon:** measure drift 1 to 4 weeks after filing rather than at quarterly resolution.

13. Stack, quality and reproduction

- **Stack:** Python 3.12, FastAPI, pydantic v2, SQLite, pandas/numpy, edgartools, the anthropic and openai SDKs (Claude or Kimi) and yfinance, plus React 19, TypeScript and Vite with a hand-rolled router and SVG charts.
- **Tests:** 34 passing, covering point-in-time cutoff enforcement, quarter rollover, transaction costs and duplicate-filing skipping. **Reproduction:** the processed buffer and price files ship in the repository, and `docker compose up --build` serves the results in this document with no API key.